

Unit - 9(Biotechnology and Its Applications) - 200 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



PART-1 (SECTION-A)

1. Which of the following is a biotechnological product-

- (1) Vaccine
- (2) Enzymes
- (3) Cheese
- (4) All of these





2. Making multiple copies of any template DNA in a host cell is known as

- (1) Replication
- (2) Transcription
- (3) Reverse transcription
- (4) Cloning





3. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : EcoRI comes from Escherichia coli RY13

Reason (R) : In EcoRI the letter 'R' is derived from the name of statin

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





4. Stanley Cohen isolated the antibiotic resistance gene and linked it to a native plasmid of

- (1) *Salmonella typhimurium*
- (2) *Salmonella pneumoniae*
- (3) *Saccharomyces cerevisiae*
- (4) *E. coli*





5. Molecular scissors are-

- (1) Helicases
- (2) Restriction endonucleases
- (3) Ligases
- (4) Topoisomerases





6. Which of the following is a vector

- (1) Plasmid
- (2) Cosmid
- (3) Phagemid
- (4) All of these





7. The new DNA molecule generated by the linking of foreign DNA to vector is called

- (1) Plasmid DNA
- (2) Recombinant DNA
- (3) Combination DNA
- (4) Chromosomal DNA





8. The first restriction endonuclease isolated was

- (1) BamH I
- (2) Sal I
- (3) Hind II
- (4) EcoR I





9. Hind II cuts DNA at a specific sequence of

- (1) 5 base pairs
- (2) 7 basepairs
- (3) 6 base pairs
- (4) 8 base pairs





10. In the nomenclature of restriction endonuclease enzyme the first letter and the successive two letters denote

- (1) Strain, Genus
- (2) Genus, strain
- (3) Genus, species
- (4) Strain, species





11. EcoR I is isolated from

- (1) E.coli RY11
- (2) E.coli RY13
- (3) E.coli RY10
- (4) E.coli RY15





12. What is correct about Hind-II

- (1) It is the first restriction endonuclease
- (2) Its functioning depends on a specific DNA nucleotide sequence
- (3) It cuts DNA by identifying a specific sequence of six base pairs
- (4) All of these





13. The specific sequence recognized by endonuclease is called-

- (1) Parallel sequence
- (2) Palindromic nucleotide sequence
- (3) Polynucleotide sequence
- (4) None of these





14. The fragments of DNA produced by restriction digestion are separated by

- (1) Chromatography
- (2) Gel electrophoresis
- (3) Centrifugation
- (4) None of these





15. Given below are two statements :

Statement I : In the year 1963, the three enzymes responsible escherichia coli were isolated.

Statement II : The first restriction endonuclease -Hind II Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





16. Under an electric field DNA moves towards :

- (1) Anode
- (2) Cathode
- (3) Don't move
- (4) None





17. Negative charge of DNA is due to the presence of

- (1) Hydroxyl group
- (2) Phosphate group
- (3) Amino group
- (4) All of these





18. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Alpha- lactalbumin was nutritionally a less balanced product to human baby.

Reason (R) : First transgenic cow, Rosie, produces human protein-enriched milk.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





19. In gel electrophoresis DNA fragments separate according to their

- (1) Size
- (2) Number of DNA fragments
- (3) Only cathode
- (4) None of these.





20. Which of the following is the matrix of DNA gel electrophoresis

- (1) Starch
- (2) Alum
- (3) Agarose
- (4) Cesium chloride





21. Agarose is extracted from

- (1) E.coli
- (2) S.typhimurium
- (3) Yeast
- (4) Sea weeds





22. Agarose is chemically a

- (1) Monomer
- (2) Dimer
- (3) Polymer
- (4) Oligomer





23. In agarose gel electrophoresis the DNA fragment which moves farthest from loading place

- (1) Largest
- (2) smallest
- (3) no information for size can be given
- (4) medium size





24. The DNA fragments separated by electrophoresis can be visualized by staining with

- (1) haematoxylin
- (2) Geimsa
- (3) Gram's stain
- (4) Ethidium bromide





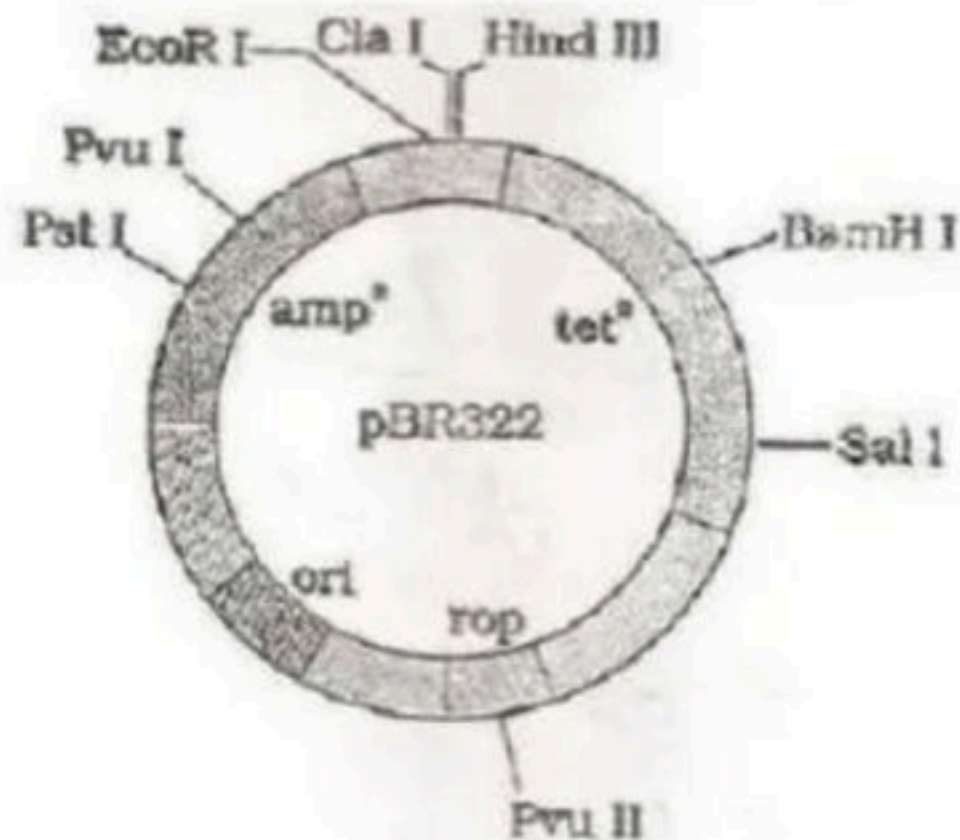
25. DNA stained with ethidium bromide can be visualized in

- (1) Visible light
- (2) UV light
- (3) IR light
- (4) X ray





26. The given below figure is the diagrammatic representation of the E.Coli vector pBR 322. Which one of the given options correctly identifies its certain component (s):



- (1) Hind III, EcoRI - selectable markers
- (2) amp^r, tet^r - antibiotic resistance genes
- (3) ori-original restriction enzyme
- (4) rop - reduced osmotic pressure





27. Process by which E. coli take up foreign DNA

- (1) Transcription
- (2) Translation
- (3) Transformation
- (4) Transfer





28. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Very low concentration of a bacteria or virus can be detected by amplification of their nucleic acid.

Reason (R) : ELISA based on the principle of antigen-antibody interaction.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





29. pBR322 is a

- (1) Natural plasmid
- (2) Artificial vector
- (3) Have ampicillin and tetracycline resistance genes
- (4) Both 2 and 3





30. *A. tumefaciens* deliver a specific piece of DNA in normal plant cells

- (1) F DNA
- (2) A DNA
- (3) T- DNA
- (4) K-DNA





31. The *agrobacterium tumifaciens*, is not a pathogen of :

- (1) Several algae
- (2) All dicot plants
- (3) All gymnosperms
- (4) All of these





32. Competent cells are made by treating bacterial cells with

- (1) Sucrose
- (2) Divalent cation
- (3) NaCl
- (4) Cold water





33. Method in which foreign DNA is directly injected in the nucleus

- (1) Micro-injection
- (2) Gene gun
- (3) Transformation
- (4) None of these





34. Which of the following are the methods of Foreign gene transfer in host cell

- (1) Biolistic
- (2) Gene gun
- (3) Disarmed pathogen vector
- (4) All of these





35. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Recombinant DNA are the DNA of two sources

Reason (R) : Restriction endonucleases are used in genetic engineering to form recombinant.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





PART-1 (SECTION-B)

36. An example of disarmed pathogen vector is

- (1) pBR322
- (2) pUC
- (3) modified Ti plasmid
- (4) YAC





37. Cell wall of bacteria can be lysed by

- (1) Lysozyme
- (2) Cellulase
- (3) Chitinase
- (4) Lignase





38. Plant cell wall can be degraded by the treatment of

- (1) Lysozyme
- (2) Cellulase
- (3) Chitinase
- (4) Lignase





39. RNA and protein portion are removed during DNA isolation by treating with

- (1) Lysozyme , cellulose
- (2) Chitinase, protease
- (3) RNase, protease
- (4) RNA polymerase, protease





40. Isolated and purified DNA can be precipitated by adding

- (1) Chilled NaCl
- (2) Chilled sucrose
- (3) Chilled ethanol
- (4) Chilled glucose





41. Amplification of gene of interest is done using

- (1) Gel electrophoresis
- (2) Chromatography
- (3) Polymerase chain reaction
- (4) Gene sequencing





42. Given below are two statements :

Statement I : The greater the BOD of waste water, more is its polluting potential.

Statement II : The remaining major of sludge is pumped into large tanks called aerobic sludge digesters.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





43. In Polymerase chain reaction there are three main steps which are repeated in several cycles. What is the correct sequence

- (1) Denaturation, extension and annealing
- (2) Denaturation, annealing and extension
- (3) Annealing, denaturation and extension
- (4) Annealing, extension and denaturation





44. In polymerase chain reaction during denaturation step

- (1) Primer is added to the two strands of DNA separated
- (2) The two strands of DNA are separated due to heat treatment
- (3) The DNA polymerase adds nucleotides to primer
- (4) Restriction endonuclease cuts DNA at recognition site





45. The primers used during annealing step of PCR are

- (1) Naturally occurring oligonucleotides
- (2) Chemically synthesized oligonucleotides
- (3) Large RNA molecule
- (4) Small tRNA molecules





**46. Thermostable DNA polymerase (Taq polymerase)
is isolated from**

- (1) E.coli
- (2) S. typhimurium
- (3) Thermus aquaticus
- (4) S. pneumoniae





47. DNA is a :

- (1) Hydrophilic molecule
- (2) Hydrophobic molecule
- (3) Lyophilic molecule
- (4) None of these





48. In continuous culture cells are maintained in

- (1) lag phase
- (2) log phase
- (3) stationary phase
- (4) none of these





49. For the transfer of foreign DNA cells are bombarded with high velocity of microparticles coated with DNA in a method known as :

- (1) Biolistics
- (2) Gene gun
- (3) Gel electrophoresis
- (4) Both 1 and 2





50. In gene gun method high velocity microparticles made up of :

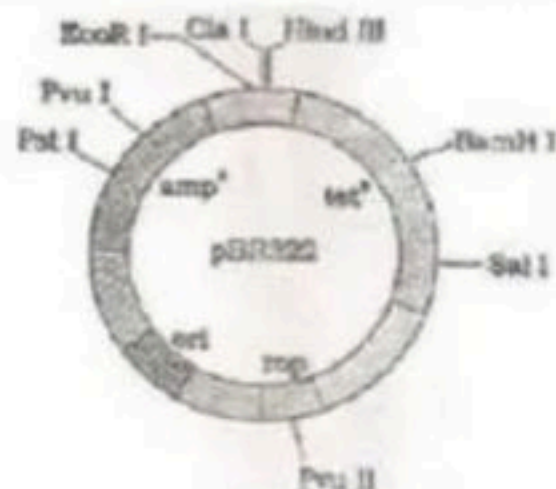
- (1) Gold
- (2) Tungsten
- (3) Aluminium
- (4) Both 1 and 2





PART-2 (SECTION-A)

51. The given below figure is the diagrammatic representation of the E.Coli vector pBR 322. Which one of the following are correct statements with reference to the given diagram :



(a) If an alien DNA ligate at BamH I site then ampicillin work as selectable marker

(b) If an alien DNA ligate at Pst I site then tetracycline work as selectable marker.

(c) If an alien DNA ligate at Sal I site then recombinants will grow in ampicillin containing medium.

- (1) a, b, c
- (2) Only a and c
- (3) Only c
- (4) Only a, b





52. Which one is vectorless gene transfer method :

- (1) Microinjection
- (2) Biolistic
- (3) Genegun
- (4) All of these





53. Presence of more than one recognition site in vector DNA :

- (1) It is beneficial process of gene cloning because more than one attachment sites are available
- (2) Complicates the gene cloning process because so many fragments are generated
- (3) Is very commonly observed in vectors
- (4) Both 1 and 3





54. We know that in PCR replication of DNA is repeated many times, the segment of DNA can be amplified to approximately billion times i.e., 1 billion copies are made in 30 cycles. Now if we make 16 billion copy then how many cycles are required :

- (1) 120
- (2) 150
- (3) 33
- (4) 34





55. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : DNA is transferred into E.coli, It is called transtormation

Reason (R) : DNA polymerase enzyme are using to make multiple copies.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





56. A gene whose expression helps to identify transformed cell is known as :

- (1) Vector
- (2) Plasmid
- (3) Structural gene
- (4) Selectable marker





57. Which one of the following is responsible for controlling of copy number of the linked DNA :

- (1) Ori
- (2) Selectable marker
- (3) Cloning site
- (4) Rop





58. Which of the following is included a component of downstream processing :

- (1) Separation
- (2) Purification
- (3) Preservation
- (4) All of these





59. In gel electrophoresis sieving effect are related with :

- (1) Size of DNA fragment
- (2) Purines bp are present in DNA fragment
- (3) Pyrimidines bp are present in DNA fragments
- (4) All of these





60. What is the meanings of the p, B, R in pBR322 respectively :

- (1) Plasmid, Bacillus, Radioactive
- (2) Plasmid, Bacillus, Rodriguez
- (3) Plasmid, Bolivar, Rodriguez
- (4) Plasmid, Bolivar, Radioactive





61. If any protein encoding gene is expressed in a heterologous host, it is called :

- (1) Special protein
- (2) Heavy protein
- (3) Recombinant protein
- (4) All of these





62. Stirred-tank bioreactors have been designed for

- (1) Purification of product
- (2) Addition of preservatives to the product
- (3) Availability of oxygen throughout the process
- (4) Ensuring anaerobic conditions in the culture vessel.





63. Which of the following is not a feature of the plasmids :

- (1) Transferable
- (2) Single-stranded
- (3) Independent replication
- (4) Circular structure





64. The DNA molecule to which the gene of interest is integrated for cloning is called :

- (1) Template
- (2) Non transformants
- (3) Transformants
- (4) Vector





65. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : The first restriction endonuclease is - Hind III

Reason (R) : DNA nucleotide sequence was isolated and characterised sequence five year later.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





66. Which one of the following restriction enzyme have recognition sequence of six base pairs:

- (1) EcoRI
- (2) Hind II
- (3) Both 1 and 2
- (4) None of these





67. In pBR322 what is the meaning of 322 :

- (1) A quality Numeral number
- (2) A numeral number which distinguish from it other developed plasmid
- (3) A stability numeral number
- (4) All of these





68. Which of the following restriction enzymes produces blunt ends :

- (1) Sal I
- (2) EcoRV
- (3) Xho I
- (4) Hind III





69. The total restriction site are present in pBR322

- (1) 6
- (2) 4
- (3) 8
- (4) 5





70. Comparing asexual reproduction with sexual reproduction, which among the following is incorrect :

- (1) Former provide opportunities for variations and formulation of unique combination of genetic set – up
- (2) Later provides opportunities for variations and formulation of same combination of genetic set – up, which are always beneficial to the organism as well as population
- (3) Sexual reproduction preserves genetic information
- (4) All of these





71. Given below are four statements pertaining to separation of DNA fragments using gel electrophoresis. Identify the incorrect statements.

i. DNA is positively charged molecule and so it is loaded on gel towards the anode terminal.

ii. DNA fragments travel along the surface of the gel whose concentration does not affect movement of DNA

iii. Smaller the size of DNA fragment larger is the distance it travels through it .

iv. Pure DNA can be visualized directly by exposing UV radiation.

(1) i, iii and iv

(2) i, ii and iii

(3) ii, iii and iv

(4) i, ii and iv





72. Downstream processing of drug include

- (1) Separation
- (2) Purification
- (3) Clinical trials
- (4) All of these





73. What is correct about bio-reactors

- (1) These are vessels in which biological products are chemically converted into specific products
- (2) These are vessels in which chemical products are converted biologically into raw materials
- (3) These are vessels in which raw materials are biologically converted into specific raw materials
- (4) These are vessels in which result in (biologically) formation of specific products from raw materials





74. While isolating DNA from bacteria, which of the following enzymes is not used

- (1) Lysozyme
- (2) Ribonuclease
- (3) Deoxyribonuclease
- (4) Protease





75. Given below are two statements :

Statement I : ELISA is based on the principle of antigen-Antibody interaction.

Statement II : PCR is now routinely used to detect HIV in suspected AIDS patients.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





76. A bioreactor provides optimum growth condition. Which among the following is included in those conditions

- (1) Temperature, Oxygen
- (2) Substrate, Salt
- (3) Vitamin
- (4) All of these





77. Which association group given the definition of biotechnology :

- (1) FBI
- (2) NACO
- (3) EFB
- (4) FIFA





78. Which one of the following is not a type of enzyme:

- (1) Cellulase
- (2) Chitinase
- (3) Lysozyme
- (4) None of these





79. Which of the following are palindromic sequence:

(1) 5' – GAATTC – 3'

3' – CTTAAG – 5'

(2) 5' – TATATA – 3'

3' – ATATAT – 5'

(3) 5' – GAGCTC – 3'

3' – CTCGAG – 5'

(4) All of these





80. Which among these are known as the main techniques which are responsible for the origin of modern biotechnology

- (a) Genetic engineering**
- (b) Phycology**
- (c) Antibiotics**
- (d) Creation of sterile ambience**

- (1) a, b**
- (2) b, c**
- (3) a, d**
- (4) a, b, c, d**





81. To overcome unwanted results of traditional hybridization procedures what technique/s was/ were useful

- (1) Only recombinant DNA
- (2) Only gene cloning
- (3) Only gene transfer
- (4) All of these





82. What is the fate of a piece of DNA which is somehow transferred into alien organism

- (1) It will multiply itself in the progeny cells of the organism
- (2) It will take control of the host genetic machinery
- (3) It will be not integrating with the genome of the recipient so that it can not co-ordinate with 'ori' sequence to start replication
- (4) Most likely it will not be able to multiply itself in progeny cells of organism





83. Accomplishment of first recombinant DNA could be possible in which year

- (1) 1972
- (2) 1971
- (3) 1973
- (4) 1975





84. Besides first restriction endonuclease how many restriction enzymes are known today

- (1) 230
- (2) 900
- (3) More than 900
- (4) Nearly 900





85. In Eco RI, what is 'R'

- (1) Roman name of the gene
- (2) Roman number
- (3) Name of the strain of DNA
- (4) Name of the strain of bacteria





PART-2 (SECTION-B)

86. Restriction enzymes belong to a larger class of enzymes called as

- (1) Nucleases
- (2) Exonucleases
- (3) Endonucleases
- (4) All of these





87. What is the functional procedure of restriction endonuclease

- (1) Identifying specific DNA sequence, cutting one strand of DNA double helix in the sugar phosphate backbone
- (2) Identifying specific DNA sequence, binding to DNA, cutting each of the two strands of single helix
- (3) Identifying specific DNA sequence, binding to DNA, cutting of each strand of DNA double helix in at nitrogen base bonds only
- (4) Identifying specific recognition sequence, binding to DNA, cut each of the two strands of the double helix at specific points in sugar-phosphate backbone





88. Regarding given figure what is the appropriate statement



- (1) Lane 1 is showing migration of undigested DNA fragment
- (2) Lane 2-4 is showing digested set of DNA fragments
- (3) All the lanes show digested set of DNA fragments
- (4) 1 and 2





89. Eco RI cuts DNA between bases G and A only when

- (1) AAGTTC is present in DNA
- (2) GATTC is present in DNA
- (3) GAATTC is present in ribonucleic acid
- (4) GAATTC is present in DNA





90. Which of the following restriction sites are not present on pBR322

- (1) Pst-I, Eco RI
- (2) Cla-I, Hind-III
- (3) Hind-III, Pst-I
- (4) Hind-II, Pst-II





91. Given below are two statements :

Statement I : The puffed up appearance of dough is due to the production of NO₂ gas.

Statement II : The LAB play very beneficial role in checking disease causing microbes.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





92. Elution step involves which of the following from the given here

- (1) Gel electrophoresis
- (2) Cutting out of the DNA bands from agarose gel
- (3) Extraction of DNA from gel piece
- (4) 2 and 3





93. Usually which one of the following is not considered as useful selectable marker in E.Coli

- (a) Ampicillin**
 - (b) Kanamycin**
 - (c) Chloramphenol**
 - (d) Chloramphenicol**
- (1) c (2) a,d
(3) d, b (4) All of these





94. An enzyme catalysing the removal of nucleotides from the ends of DNA is:

- (1) Endonuclease
- (2) Exonuclease
- (3) DNA ligase
- (4) Hind - II





95. The most important feature in a plasmid to be used as a vector is:

- (1) Origin of replication (ori)
- (2) Presence of a selectable marker
- (3) Presence of sites for restriction endonuclease
- (4) Its size





96. Who has got the ability to transform normal cells into cancerous cells in animals :

- (1) Retrovirus
- (2) Reovirus
- (3) Rhabdovirus
- (4) Rhinovirus





97. In order to make bacterial cell competent to take recombined DNA, these are treated with divalent cations such as calcium. How this enables the entry of recombinant DNA into the cell

- (1) It creates gradient
- (2) This increases the efficiency with which DNA enters the bacterium through pores in its cell wall
- (3) It causes active uptake of DNA by the cell
- (4) It causes active uptake of RNA by the cell





98. What is required during the extension step of polymerase chain reaction

- (a) DNA polymerase**
- (b) Taq polymerase**
- (c) Deoxynucleosides**
- (d) Oligonucleotides**

- (1) a, b**
- (2) a, b, c only**
- (3) b, c, d**
- (4) All of these**





99. The process of annealing requires

- (1) One set of small chemically synthesized oligonucleosides
- (2) Two set of primers (large chemically synthesized oligonucleotides)
- (3) Two sets of primers (small chemically synthesized oligonucleotides)
- (4) All of these





100. For obtaining appreciable quantities of products through biotechnology how much culture can be processed in bioreactors :

- (1) 100 – 1000 litres
- (2) 1000 – 10000 litres
- (3) 500 litres
- (4) 10 – 20 litres





PART-3 (SECTION-A)

101. Which of the following genes were introduced in cotton to protect it from cotton bollworms ?

- (1) Cry Ac and Cry Ab
- (2) Bt Ac and Bt Ab
- (3) Cry IAc and Cry II Ab
- (4) Nif genes





102. 'Golden rice' developed through transgene approach is enriched with:

- (1) High lysine content
- (2) High methionine content
- (3) High glutenin content
- (4) High vitamin - A content





103. Agro-chemical based agriculture includes

- (1) Fertilizers and pesticides
- (2) Genetically modified crops
- (3) RNA interference
- (4) All of these





104. Bt corn has been made resistant from corn borer disease by introduction of the gene

- (1) Cry I Ab
- (2) Cry II Ab
- (3) ampR
- (4) Trp





105. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : DNA fragments can be separated by gel electrophoresis.

Reason (R) : DNA fragments are negatively charged molecules they can be separated by forcing them to move towards the anode under an electric field through a medium.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





106. Given below are two statements :

Statement I : The toxin is coded by gene cryIAc named cry.

Statement II : RNAi takes place in all prokaryotic organisms as a method of cellular defense.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





107. Bt toxin kills insects by :

- (1) Inhibiting protein synthesis
- (2) Generating excessive heat
- (3) Creating pores in the midgut epithelial cells, leading to cell swelling and lysis
- (4) Obstructing a biosynthetic pathway





108. RNA interference involves :

- (1) Synthesis of cDNA and RNA using reverse transcriptase
- (2) Silencing of specific mRNA due to complementary dsRNA
- (3) Interference of RNA in synthesis of DNA
- (4) Synthesis of mRNA from DNA





109. Which is correct :

- (1) Transgenic animals are used in study of complex factors involved in growth such as insulin – like growth factor
- (2) Transgenic models are not used for study of diseases like cancer and Alzheimers disease
- (3) β -5-antipepsin is used to treat emphysema
- (4) Transgenic dog are being used to test the safety of polio vaccine.





110. Which of the following statements regarding the structure of proinsulin and mature insulin are not correct ?

- (i) Proinsulin is made up of three polypeptide chains – A, B and C
- (ii) C – peptide is not present in the mature insulin
- (iii) Mature insulin is made up of 51 amino acids arranged in two polypeptide chains – A and B
- (iv) Polypeptide chain A has 30 amino acids and polypeptide chain B has 21 amino acids
- (v) Polypeptide chains A and B are interconnected by only one S – S linkage

- | | |
|------------------|-------------------------|
| (1) (i) and (ii) | (2) (iii) and (iv) |
| (3) (iv) and (v) | (4) (iii), (iv) and (v) |





111. Early detection of a disease is possible by :

- (1) PCR
- (2) ELISA
- (3) Recombinant DNA technology
- (4) All of these





112. ADA deficiency can be permanently cured by which of the following methods ?

- (1) Bone marrow transplantation
- (2) Enzyme replacement therapy
- (3) Gene therapy at early embryonic stages
- (4) All of these





113. It is a single stranded DNA or RNA, tagged with a radioactive molecule and is used to detect mutated genes.

- (1) RNAi
- (2) Probe
- (3) Plasmid
- (4) Primer





114. Which of the following statements is incorrect about gene therapy in ADA deficiency ?

- (1) Lymphocytes from patient's blood are taken out and cultured
- (2) A functional ADA cDNA is introduced into these lymphocytes
- (3) Lymphocytes are then introduced in the body of patient
- (4) Patient does not require periodic infusion of genetically engineered lymphocytes





115. During the processing of proinsulin into the mature insulin

- (1) C - peptide is added to proinsulin
- (2) C - peptide is removed from proinsulin
- (3) B - peptide is added to proinsulin
- (4) B - peptide is removed from proinsulin





116. Animlas that have their DNA manipulated to possess and express a foreign gene are called :

- (1) Transgenic animals
- (2) Somatic hybrids
- (3) Somaclones
- (4) Super animals





117. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Restriction endonuclease are used in genetic engineering to form recombinant molecules of DNA.

Reason (R) : The recombinants will grow in ampicillin containing medium if pBR³²⁰ integrated on tet^R gene.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





118. Match the column - I with column - II and select the correct answer from the codes given below :

Column - I

A. α -1-antitrypsin

B. Transposon

C. ELISA

D. Retroviral vector

Column - II

(i) AIDS

(ii) Gene therapy

(iii) Emphysema

(iv) Mobic genetic element

(1) A - (i), B - (iii), C - (ii), D - (iv)

(2) A - (iii), B - (iv), C - (i), D - (ii)

(3) A - (i), B - (ii), C - (iii), D - (iv)

(4) A - (iii), B - (i), C - (ii), D - (iv)





119. Match column - I containing transgenic organisms with their specific characteristics in Column – II and select the correct answer from codes given below :

Column - I

- A. Golden rice**
- B. Bt cotton**
- C. Flavr Savr**
- D. Rosie cow**

Column - II

- (i) Protein - enriched milk**
- (ii) Increased self life**
- (iii) Enriched with vitamin A**
- (iv) High yield and pest resistant**

- (1) A - (iii), B - (iv), C - (ii), D - (i)**
- (2) A - (iii), B - (ii), C - (iv), D - (i)**
- (3) A - (ii), B - (iv), C - (iii), D - (i)**
- (4) A - (i), B - (iv), C - (ii), D - (iii)**





120. Use of bio-resources by multinational companies and other organizations without proper authorisation from the countries and people concerned without compensatory payment is termed as :

- (1) Resource partitioning
- (2) Biopiracy
- (3) Patenting
- (4) Biofortification





121. Match Column - I with Column - II and select the correct answer from the codes given below :

Column - I

A. Biopiracy

B. Biopatent

C. Gene therapy

D. RNAi

Column - II

(i) Effort to fix the non-functional gene

(ii) Gene silencing

(iii) Illegal removal of biological materials

(iv) Right granted for biological entities

(1) A - (iv), B - (ii), C - (i), D - (iii)

(2) A - (ii), B - (iv), C - (i), D - (iii)

(3) A - (iii), B - (iv), C - (i), D - (ii)

(4) A - (iii), B - (iv), C - (ii), D - (i)





122. Which variety of rice was patented by a U.S. company even though the highest number of varieties of this rice are found in India ?

- (1) Sharbati Sonora
- (2) Co-667
- (3) Basmati
- (4) Lerma Rojo





123. Bt cotton is not :

- (1) A GM plant
- (2) Insect resistant
- (3) A bacterial gene expressing system
- (4) Resistant to all pesticides





124. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) :

Nucleic acid is the genetic material of all organism with some exception.

Reason (R) :

Majority of organism this is deoxyribo nucleic acid or DNA.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





125. GEAC stands for :

- (1) Genome Engineering Action Committee
- (2) Ground Environment Action Committee
- (3) Genetic Engineering Approval Committee
- (4) Genetic and Environment Approval Committee





126. The site of production of ADA in the body is :

- (1) Erythrocytes
- (2) Lymphocytes
- (3) Blood plasma
- (4) Osteocytes





127. Golden rice is :

- (1) A variety of rice grown along the yellow river in China
- (2) Long stored rice having yellow colour tint
- (3) Enhanced nutritional value of food.
- (4) Wild variety of rice with yellow coloured grains





128. ADA is an enzyme which is deficient in a genetic disorder SCID. What is the full form of ADA :

- (1) Adenosine deoxy aminase
- (2) Adenosine deaminase
- (3) Aspartate deaminase
- (4) Arginine deaminase





129. Find out the correct statements :

- a. Most Bt toxins are insect group specific**
- b. Lepidopterans do not include beetles**
- c. GM plants helped to reduce post harvest loss**
- d. GMOs include plants, bacteria, fungi animals**

(1) a, b only

(2) b, c only

(3) a, b, c and d

(4) c and d only





130. Nematode specific genes were introduced into host plant by using :

- (1) *Bacillus thuringiensis*
- (2) *Agrobacterium* vectors
- (3) *Meloidogyne incognita*
- (4) *E. coli*





131. *Bacillus thuringiensis* (Bt) strains have been used for designing novel :

- (1) Bio-metallurgical technique
- (2) Bio insecticidal plants
- (3) Bio minerallization process
- (4) Bio fertilizer





132. Insulin used for Diabetes patients was earlier extracted from :

- (1) Pigs
- (2) Slaughtered Cattles
- (3) Both (1) and (2)
- (4) None





133. The number of documented basmati rice variety grown in India are :

- (1) 7
- (2) 27
- (3) 37
- (4) 57





134. What happens when a nematode specific gene is introduced into the host plant :

- (1) It produce only anti – sense RNA in host plant which can not perform translation hence protein required for nematode is not synthesized
- (2) Synthesis of ds RNA which silence the m – RNA of the nematode
- (3) Both 1 and 2
- (4) None





135. Given below are two statements :

Statement I :

Transformation is a procedure through which a piece of DNA is introduced in host bacterium

Statement II :

Transformation is also applicable to viruses.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect**
- (2) Statement I is correct but Statement II is incorrect**
- (3) Statement I is incorrect but Statement II is correct**
- (4) Both, Statement I and Statement II are correct**





PART-3 (SECTION-B)

136. Genes of interest can be selected from a genomic library by using :

- (1) Restriction enzymes
- (2) Cloning vectors
- (3) DNA probes
- (4) Gene targets





137. GM plants have been useful in many ways. Which among the following does not represent the advantage of GM organisms :

- (1) Made crop tolerant to abiotic stress
- (2) Reduced reliance on pesticides
- (3) Decreased efficiency of mineral usage by plants
- (4) Enhanced nutritional value





138. What are the possible permanent treatment of SCID

- (1) Enzyme replacement
- (2) Bone marrow transplant
- (3) Gene therapy of lymphocytic DNA
- (4) None





139. is now routinely used to detect HIV in suspected AIDS patients :

- (1) Radioisotope
- (2) PCR
- (3) Chromatography
- (4) Chloride test.





140. Which of the following statements is correct about Bt toxin ?

- (1) Bt protein exists as active toxin in the Bacillus
- (2) The activated toxin enters the ovaries of the pest to sterilize it and thus prevent its multiplication
- (3) The concerned Bacillus has antitoxins
- (4) The inactive protoxin gets converted into active form in the insect gut





141. Find out the incorrect statement

- (1) Human protein used to treat emphysema is α -1 antitrypsin
- (2) Human insulin is being commercially produced from a transgenic species of *Agrobacterium tumefaciens*
- (3) Rosie, the first transgenic cow, produced human protein enriched milk
- (4) Cry I Ab endotoxins obtained from *Bacillus thuringiensis* is effective against corn borers





142. Which one fo the following statement(s) is/are correct about (GEAC) ?

- (1) It will make decision regarding the validity of GM research
- (2) It will make the safety of introducing GM - organism for public services
- (3) Both (1) and (2)
- (4) None of these





143. Which of the following statements is correct :

- (1) The current interest in the manipulation of microbes, plants and animal has raised serious ethical issues
- (2) One possible risk of genetic engineering is the accidental production of dangerously resistant microorganisms
- (3) Both (1) and (2)
- (4) None of these





144. The proteins produced by *B. thuringiensis* that kill insects like :

- (1) *Lepidopterans*
- (2) *Dipterans*
- (3) *Coleopterans*
- (4) All of these





145. Bt toxin is coded by a gene named :

- (1) shy
- (2) try
- (3) cry
- (4) pry





146. Human insulin is being commercially produced from a transgenic species of :

- (1) *Mycobacterium*
- (2) *Rhizobium*
- (3) *Sacchromyces*
- (4) *Escherichia*





147. Match the items given in column - I with those in column - II and choose the correct option.

Column - I

A. GMO

B. Flavr - Savr tomato

C. Biopiracy

D. E.coli

Column - II

I. Increased shelf life

II. Bioresources

III. rDNA

IV. Insulin

(1) A - III; B - I; C - II; D - IV

(2) A - II; B - I; C - III; D - IV

(3) A - II; B - III; C - I; D - IV

(4) A - IV; B - I; C - II; D - III





148. Match the column - I with column - II and identify the correct option :

Column - I

A. Gene therapy

B. Biofertilizer

C. Bt cotton

D. Humulin

Column - II

I. Rhizobium

II. Cry gene

III. SCID

IV. Diabetes

(1) A - II; B - I; C - IV; D - III

(2) A - III; B - I; C - II; D - IV

(3) A - II; B - III; C - I; D - IV

(4) A - IV; B - I; C - II; D - III





149. Match the column - I with column - II and choose the correct option

Column - I

Column - II

A. Golden rice

I. Cry protein

B. Bt toxin

II. Rich in vitamin A

C. RNAi

III. First transgenic cow

D. Rosie

IV. Gene silencing

(1) A - II; B - I; C - IV; D - III

(2) A - II; B - I; C - III; D - IV

(3) A - II; B - III; C - I; D - IV

(4) A - IV; B - I; C - II; D - III





150. Match column - I with column - II and choose the correct option.

Column - I

A. Forensic science

B. ELISA

C. Probe

D. α -1-antitrypsin

Column - II

I. AIDS

II. Radioactive DNA/RNA

III. Emphysema

IV. DNA fingerprinting

(1) A - II; B - I; C - IV; D - III

(2) A - II; B - I; C - III; D - IV

(3) A - II; B - III; C - I; D - IV

(4) A - IV; B - I; C - II; D - III





PART-4 (SECTION-A)

151. Transgenic animals has been successfully used for producing

- (1) transgenic mice for testing safety of polio vaccine before use in humans
- (2) Transgenic models for studying new treatments for certain cardiac diseases
- (3) Transgenic cow - rosie which produces high fat milk for making ghee
- (4) Animals like bulls for farm work as they have super power





152. In 1997, the first transgenic cow Rosie produced human protein enriched milk. The protein present and the amount of the protein is :

- (1) α Lactalbumin, 2.4 gm / lit
- (2) α Lactalbumin, 1.4 gm / lit
- (3) α Lactalbumin, 12.4 gm / lit
- (4) β Lactaglobulin, 12.4 gm / lit





153. Important objective of biotechnology in agriculture sections is :

- (1) To produce pest resistant varieties of plants
- (2) To increase the nitrogen content
- (3) To decrease the seed number
- (4) To increase the plant weight





154. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) :

Cry proteins are named so because they are crystal proteins (Crystallized)

Reason (R) :

Cry proteins are solubilised in acidic medium of insect midgut the release toxin core fragments after proteolytic action.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





155. Pest resistance crops have an advantage

- (1) They can resist abiotic stresses
- (2) These crops are less dependent on pesticide sprays
- (3) These crops only survive when regular pesticides are sprayed
- (4) These crops have increased nutritional value





**156. In order for gene therapy to be most effective,
gene should be inserted in :**

- (1) WBC
- (2) RBC
- (3) Stem cells
- (4) All of these





157. Bt toxin is produced by

- (1) *Thermus aquaticus*
- (2) *Bacillus thuringiensis*
- (3) *Bacillus subtilis*
- (4) *Bacillus thermoglucosidaci*





158. Protein produced by *B. thuringiensis* kills flies is:

- (1) *Lepidopterans*
- (2) *Coleopterans*
- (3) *Dipterans*
- (4) All of these





159. Bt cotton is a variety of cotton

- (1) Which is heat tolerant
- (2) Bollworm resistant
- (3) Salt resistant
- (4) Drought resistant





- (1) Virus
- (2) Bacteria
- (3) Nematode
- (4) Arthropod

- (1) Virus
- (2) Bacteria
- (3) Nematode
- (4) Arthropod





161. Meloidegyne incognitia infects which plant part

- (1) Leaf
- (2) Stem
- (3) Root
- (4) Flower





162. RNAi stands for

- (1) RNA inhibitor
- (2) RNA interference
- (3) RNA inducer
- (4) RNA initiator





163. Given below are two statements :

Statement I :

The RNA can be removed by treatment with ribonuclease protein can be removed by treatment by protease.

Statement II :

Molecules can be removed by appropriate treatment and purified DNA ultimately precipitate out after the addition of chilled ethanol.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





164. Transposons are

- (1) Elements of replication
- (2) A vaccine
- (3) An antibiotic
- (4) Mobile genetic elements





165. The source of complementary RNA in RNA interference can be

- (1) RNA viruses
- (2) Mobile genetic elements
- (3) Both 1 and 2
- (4) None of these





166. About how many recombinant therapeutics have been approved worldover

- (1) 12
- (2) 30
- (3) 10
- (4) 50





167. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : (ADA) deficiency disorder is caused due to the deletion of gene for adenosine deaminase.

Reason (R) : Gene therapy is a collection of methods that not allow correction of a gene defect that has been diagnosed in a child/embryo.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





168. Insulin was initially extracted from

- (1) Intestine of monkeys
- (2) Pancreas of slaughtered cattle and pigs
- (3) Heart of mice
- (4) Muscles of monkeys





169. Research area of biotechnology include

- (1) Providing best catalysing form of microbe or pure enzyme
- (2) Creating optimal conditins for a catalyst to act
- (3) Downstream processing technologies
- (4) All of these





170. The constituent polypeptide chains are linked to each other in insulin by

- (1) Glycosidic bond
- (2) Phosphodiester bond
- (3) Disulfide bond
- (4) Peptide bond





171. In mammals which part does proinsulin have but mature insulin molecule doesn't have

- (1) Chain A
- (2) Chain B
- (3) C peptide
- (4) B peptide





172. Which American company prepared insulin as two separate DNA sequences A and B

- (1) Elisa Little
- (2) Eric Lilly
- (3) Elisa Lilly
- (4) Eli Lilly





173. Collection of methods that allows correction of a gene defect diagnosed in a child or embryo is called

- (1) Biochemistry
- (2) Gene silencing
- (3) Gene therapy
- (4) None of these





174. First clinical gene therapy was given in which year

- (1) 1993
- (2) 1983
- (3) 1990
- (4) 1982





175. Given below are two statements :

Statement I :

Restriction enzymes belongs to a larger class of enzymes called nuclease.

Statement II :

Exo nuclease remove nucleotides from the mid of the DNA.

Choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct





176. First clinical gene therapy was given to

- (1) 40 year old man
- (2) 14 year old boy
- (3) 4 year old girl
- (4) 4 month old girl





177. Conventional diagnosis methods include

- (1) Polymerase chain reaction
- (2) Serum and urine analysis
- (3) ELISA
- (4) Recombinant DNA technology





178. ELISA stands for

- (1) Enzyme linked immunosolving assay
- (2) Enzyme linked immunosorbent assay
- (3) Ether linked immunosorbent assay
- (4) Enzyme linked immunosorting assay





179. Probe used in autoradiography is

- (1) Single stranded RNA tagged with radioactive molecule
- (2) Single stranded DNA tagged with radioactive molecule
- (3) Both 1 and 2
- (4) None of these





180. ELISA is based on the principle of

- (1) DNA replication
- (2) Translation
- (3) Antigen antibody interaction
- (4) Radioactivity





181. Maximum percentage of transgenic animals are

- (1) Monkey
- (2) Rabbit
- (3) Cow
- (4) Mice





182. Transgenic models exist for which disease

- (1) Cancer
- (2) Cystic fibrosis
- (3) Rheumatoid arthritis
- (4) All of these





183. Safety of polio vaccine is tested on which transgenic animal :

- | | |
|---------|-----------|
| (1) Pig | (2) Mice |
| (3) Cow | (4) Sheep |





184. Approximately how many varieties of rice are there in India alone

- (1) 100000
- (2) 200000
- (3) 300000
- (4) 400000





185. In 1997, an American company got patent rights on which variety of rice

- (1) Mansoori
- (2) Ram bhog
- (3) Kala namak
- (4) None of these





PART-4 (SECTION-B)

186. What advantage do traditional knowledge related to bioresource have during their commercialisation

- (1) It saves time
- (2) Saves effort
- (3) Saves expenditure
- (4) All of these





187. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Insulin used for diabetes was earlier extracted from pancreas of slaughtered cattle and pigs.

Reason (R) : In mammal, including humans, insulin is synthesised as pro-hormones.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





188. Indian parliament have recently cleared an amendment in Indian Patent Bill. Which amendment was this

- (1) First
- (2) Third
- (3) Second
- (4) None of these





189. Find out incorrect statement in the following :

(a) DNA fragments move towards which electrode in electrophoratic technique-anode

(b) *Meloidegyne incognitia* infects which plant part- stem

(c) RNAi takes place in some eukaryotes as a method of cellular defense

(d) Insulin consist of one polypeptide chains

(1) a, b and c

(2) a, b, c and d

(3) b, c and d

(4) Only c





190. Which is incorrect :

- (1) In biotechnology microorganisms are used as catalyst
- (2) Biotechnology is used in waste management
- (3) Biotechnology has no role in energy production
- (4) Downstream process are used in purifying organic compound.





191. Application of biotechnology include :

- (1) Therapeutics
- (2) Diagnostics
- (3) Genetically modified crops
- (4) All of the above





192. Mature insulin has :

- (1) Single polypeptide chains
- (2) Two polypeptides chains
- (3) Three polypeptides chains
- (4) None of these





193. Which is incorrect :

- (1) Genetically modified crops are more tolerant to cold and drought
- (2) GM crops are pest resistant
- (3) GM crops are not helpful in reducing post harvest losses
- (4) GM crops prevent early exhaustion of fertility of soil.





194. Why does this toxin not kill the bacillus in Bt cotton:

- (1) Bt toxin exist as inactive prolipid which change into another form
- (2) Bt toxin exist as inactive protoxin but after ingestion by insect at alkaline pH of the gut which solubilise the crystals, it is converted into an active form
- (3) Bacteria has specific lipids to neutralise toxin
- (4) Bacterial proteins are immune to toxin.





195. In 1997 an American company got patent rights on Basmati rice through :

- (1) Indo US Patent department
- (2) Indo US Patent and Trade organisation
- (3) US Patent and Trademark office
- (4) US patent and Trade department.





196. Insulin consist of two short chain. Chain A and B are linked together by :

- (1) Glycosidic, hydrogen bond
- (2) Peptide, nitrogen bond
- (3) Polypeptide, disulphide bond
- (4) Glycosidic, phosphodiester bond.





197. ADA deficiency can be cured by :

- (1) Bone marrow transplantation
- (2) Enzyme replacement therapy
- (3) Both (1) and (2)
- (4) Blood transfusion





198. Which is incorrect :

- (1) Transgenic models are used to study rheumatoid arthritis
- (2) In Gene therapy lymphocytes from blood of the patient are not grown in a culture outside the body.
- (3) GM plant have been useful in increasing crop yield
- (4) Genetic manipulation of microbes, plants and animals has raised serious ethical question.





199. Indian Basmati was crossed withvarieties and claimed as an invention

- (1) Long
- (2) Dwarf
- (3) Both (1) and (2)
- (4) Semi dwarf.





200. Over 95% of all existing transgenic animals are:

- (1) Cow
- (2) Dog
- (3) Mice
- (4) Sheep

